Quantum Error Mitigation

Bachelors Thesis Project Presentation

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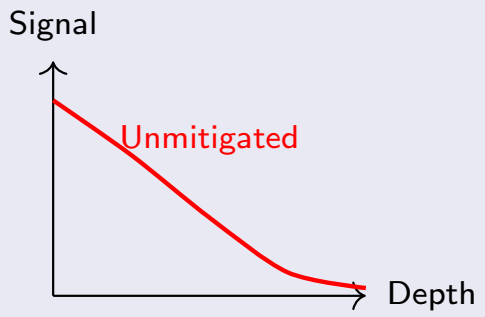
Introduction: The NISQ Bottleneck

Motivation:

- We are in the **NISQ Era** (Noisy Intermediate-Scale Quantum).
- Qubits are fragile; environmental noise (T_1, T_2) and gate errors accumulate **exponentially**.
- **The Problem:** Standard Zero-Noise Extrapolation (ZNE) requires massive runtime overhead (re-running circuits).
- We need a way to extract ideal signals with **near-zero** execution cost.

Noise Impact



Statement of Objective

Primary Goal

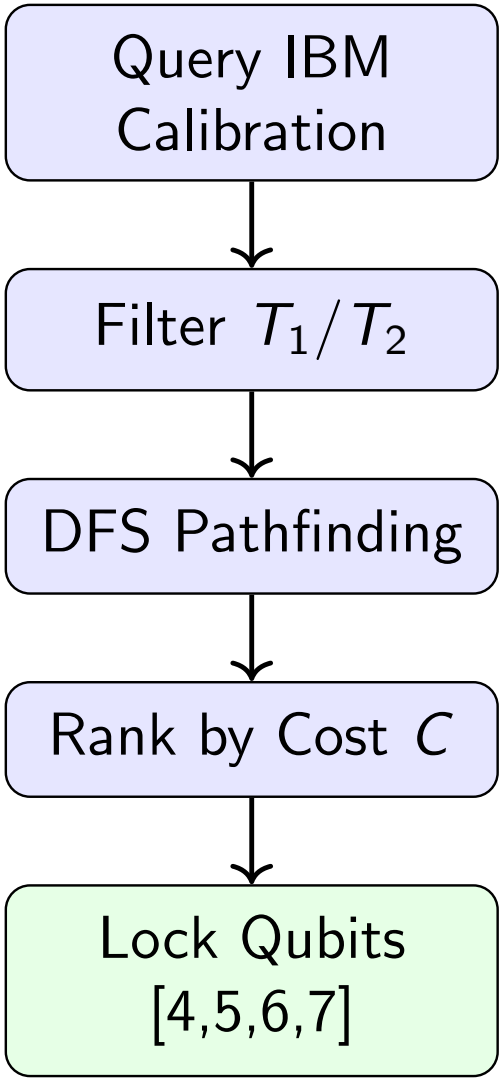
To develop an **Automated, Hardware-Aware Machine Learning Framework** that mitigates quantum errors in high-depth circuits using only a single noisy execution.

Specific Objectives:

1. **Autonomous Profiling:** Dynamically find the healthiest qubits on the IBM Cloud via graph topology.
2. **Feature Engineering:** Feed real-time calibration data (T_1, T_2, ϵ_g) into the ML model.
3. **Efficiency:** Achieve $> 70\%$ error reduction with millisecond inference time, bypassing ZNE overhead.

Methodology I: Backend Profiling

- **Graph Construction:** Model the 156-qubit heavy-hex lattice as $G = (V, E)$.
- **Filtering:** Prune qubits below the 25th percentile of coherence.
- **Topological Search:** Find all 4-qubit linear paths to avoid **SWAP gates**.
- **Cost Function:** Minimized C balancing gate error and decoherence.

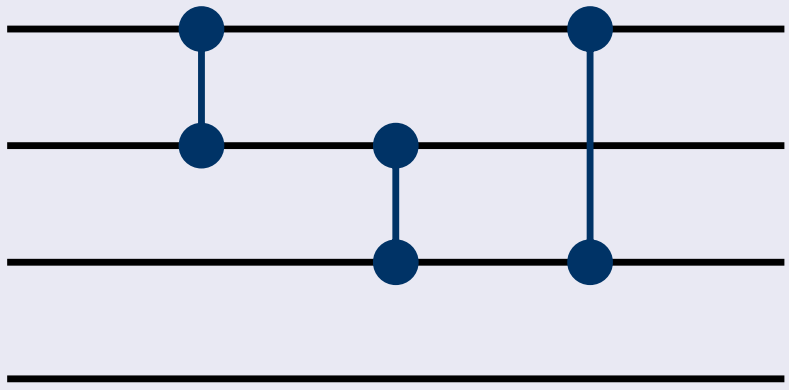


Methodology II: Circuit Design

Hong-Ou-Mandel Simulation:

- Parameterized by angle θ and Trotter depth N.
- **Depth:** Up to 178 layers.
- **Bridges:** Custom Gray Code entangling gates to minimize decoherence.
- **Transpilation:** Strict hardware locking to qubits $[4, 5, 6, 7]$ at optimization level 0.

HOM Gray Code Circuit



Repetitive Trotter Block

Methodology III: Dataset & ML Pipeline

Parallel Execution:

- **Features (X):** Noisy counts from *ibm_marrakesh* + real-time calibration.
- **Target (y):** Noiseless results from *AerSimulator*.
- **Total Data:** 45 samples (15 angles $\times$ 3 depths).

Model Comparison:

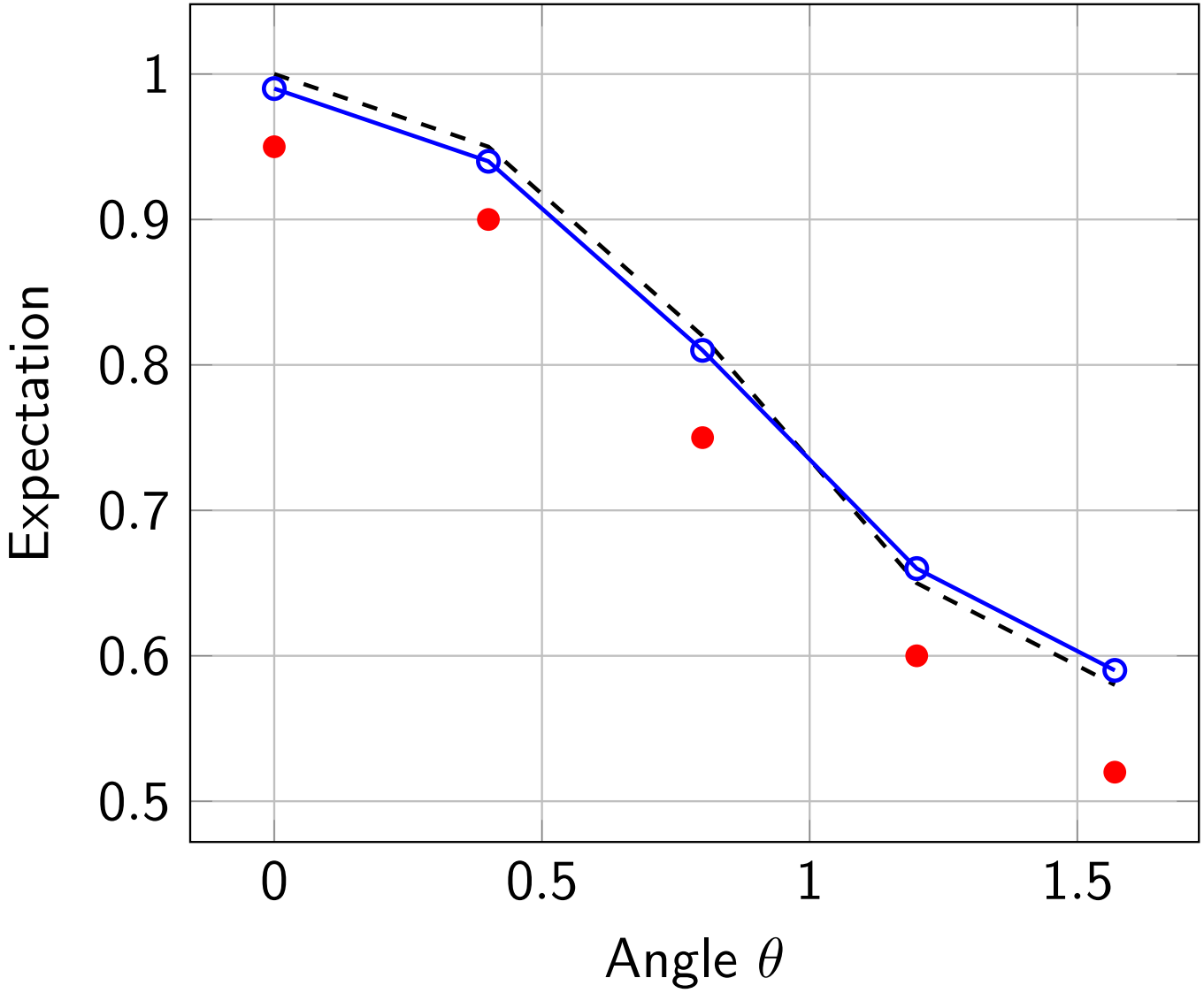
- **Linear OLS:** Benchmark to prove non-linearity.
- **Gradient Boosting:** Sequential boosting approach.
- **Random Forest:** Bagging ensemble (150 trees) for variance reduction.

Results: Mitigation Performance

Strategy	MAE	Improvement
Raw Hardware (Baseline)	**0.03326**	**0.00%**
Single Decision Tree	0.02298	30.90%
Gradient Boosting (Boosting)	0.01517	54.40%
Random Forest (Bagging)	**0.00726**	**78.17%**

- **Observation:** Random Forest achieves the highest accuracy.
- **Metric:** 78.17% reduction in Mean Absolute Error.
- **Reliability:** Error bars reduced through ensemble averaging.

Results: Physics Tracking



--- Ideal Physics ● Raw Hardware

Discussion: Drivers of Algorithmic Success

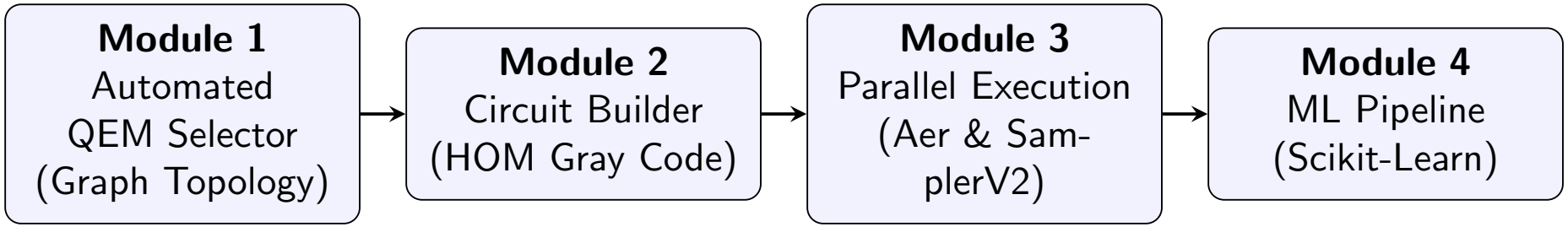
- **Why Random Forest Won:** Gradient Boosting "chased" random noise spikes. Bagging (RF) averaged 150 trees, naturally smoothing out **quantum shot noise**.
- **Feature Importance:** Including T_1 and T_2 as features allowed the model to scale its correction based on real-time chip health.
- **Computational Gain:** Mitigation happens in **milliseconds** classically. Compared to ZNE, this framework is viable for real-time quantum tasks.

Final Result: 78.2% Accuracy Improvement

Thank You!
Questions?

Backup: Software Architecture & Code Flow

How the Framework Operates:



- **Object-Oriented Design:** The code is modular, separating the IBM cloud API calls from the Machine Learning post-processing.
- **Fault Tolerance:** Implements a "Job Rescue" protocol to retrieve results if the IBM quantum cloud queues timeout during execution.

Backup: DFS Pathfinding & Finding [4,5,6,7]

The DFS Algorithm:

- *ibm_marrakesh* uses a heavy-hex lattice.
- We use `networkx.all_simple_paths(G, src, cutoff=3)`.
- **Goal:** Find all strict 4-node lines.
- **Why?** A linear sub-graph ensures native CZ gates without the compiler inserting destructive `SWAP` operations.

Why Qubits [4,5,6,7] Won

While hundreds of 4-qubit paths exist, path [4,5,6,7] uniquely possessed:

- $\epsilon_{gate} < 0.01$ (low cross-talk)
- High T_1 and T_2 coherence times compared to the chip average.
- Safely passed the 25th percentile coherence filter.

Backup: Filtering & Composite Cost Function

The Formula

$$C = \sum_{i \in E} \epsilon_{\text{gate}}^{(i)} + \sum_{q \in V} \epsilon_{\text{readout}}^{(q)} + \lambda \sum_{q \in V} \left(\frac{1}{T_1^{(q)}} + \frac{1}{T_2^{(q)}} \right)$$

Mathematical Breakdown:

- **Term 1 (Edges):** Sum of native 2-qubit entangling gate errors.
- **Term 2 (Nodes):** Sum of measurement/readout assignment errors.
- **Term 3 (Decoherence Penalty):** Inverses of T_1 and T_2. Shorter coherence times create a massive mathematical penalty.
- λ **(Hyperparameter):** A tunable weight to prioritize either gate fidelity or environmental decoherence based on circuit depth.

Backup: Feature Engineering

Why traditional ML fails for Quantum: Standard models map $X \to Y$ using only the noisy expectation. They fail when the hardware state drifts.

Hardware-Aware Features Used:

Feature	Type	Physical Significance
`noisy_exp`	Float	The raw collapsed state from the IBM QPU.
`theta` (θ)	Float	Beam splitter angle (probability amplitude).
`trotter_steps`	Int	Circuit depth (exponential error multiplier).
`avg_T1`	Float	Defines the rate of energy relaxation (decay).
`avg_T2`	Float	Defines the rate of phase loss (decoherence).

Backup: Dataset Generation Strategy

Parameter Sweep:

- We must prevent the ML from memorizing a single state.
- Swept θ from 0 to $\pi/2$ (15 linear steps).
- Swept Trotter Depths $N \in \{1, 2, 3\}$.
- Yields **45 unique circuits**.

Execution Constraints

Qiskit Transpiler Rules applied:

- `initial_layout=[4,5,6,7]`
- `optimization_level=0`

This strictly forces the compiler to use our optimal path and prevents it from rewriting our custom Gray code structure.

Backup: ML-QEM vs ZNE Comparison

Why this project matters in the NISQ landscape:

Feature	ZNE (Industry Standard)	ML-QEM (This Work)
Runtime Cost	High ($O(k)$ executions per point)	**Near-Zero** (1 execution)
Hardware Prep	Digital Gate Folding (adds depth)	**None** (uses native circuit)
Inference	Mathematical Extrapolation	**Trained Regression**
Requirement	Assumes noise scales linearly/exponentially	**Data-driven / Non-linear**

- ZNE dominates QPU execution time, making complex tasks prohibitively slow and expensive on cloud resources. ML-QEM shifts that burden to fast, classical Python processing.

Backup: Trotterization & Exponential Error

What is Trotterization?

To simulate quantum dynamics, we apply $U(t) = e^{-iHt}$. For $H = H_1 + H_2$, we approximate using the **Suzuki-Trotter expansion**:

$$e^{-i(H_1+H_2)t} \approx \left(e^{-iH_1 \frac{t}{N}} e^{-iH_2 \frac{t}{N}}\right)^N$$

- N is the **Trotter Depth**.
- Higher accuracy requires larger N.
- Each step N adds entangling gates.

Exponential Error Scaling

If a gate has error ϵ, its fidelity is $f = 1 - \epsilon$. Total fidelity for depth D is the product of all gate fidelities:

$$F_{\text{total}} = \prod_{i=1}^{D} (1 - \epsilon_i) \approx (1 - \epsilon)^D \approx e^{-D\epsilon}$$

- Circuit depth D scales linearly with N.
- Physical signal **decays exponentially**.

- **Why ML-QEM is necessary here:** Reaching Trotter Step 3 requires **178 gate layers**. At this depth, exponential decay pulls the raw signal down into the depolarizing limit (pure noise), making it incredibly difficult for standard methods to recover the physics.

Backup: Quantum Hardware Errors

1. Coherent Errors (Systematic)

Predictable, reversible deviations.

- **Over/Under-rotation:** Miscalibrated pulses (e.g., 182° instead of 180°).
- **Off-resonance:** Frequencies don't perfectly match, tilting the rotation axis.

2. Incoherent Errors (Decoherence)

Irreversible loss of quantum information.

- **Amplitude Damping (T_1):** Qubit spontaneously loses energy ($|1\rangle \to |0\rangle$).
- **Phase Damping (T_2):** Fluctuating fields destroy the superposition phase.

3. Cross-talk (Spectator Errors)

Unintended electromagnetic interactions.

- Driving a target accidentally applies a small rotation to neighbors.
- **ZZ Cross-talk:** A neighbor's state dynamically shifts your target's frequency.

4. Leakage Errors

Leaving the computational subspace.

- Physical qubits (like transmons) have higher states ($|2\rangle, |3\rangle$).
- Intense/fast pulses excite the qubit to $|2\rangle$, breaking the algorithm.